

Lecture 11: Regression

Spring 2026

10 ECTS Credits

Department of Information and Communication Technology

Faculty of Engineering and Science

Recap

- Clustering
 - K-Means Clustering
 - Hierarchical Clustering
 - Self-Organizing Maps (SOM)



Optimal Regression

- The objective is to predict the value of a target $Y \in \mathbb{R}$ given a feature vector $\mathbf{X} \in \mathbb{R}^d$.
- No single gold standard performance criterion.
- An optimal predictor can be obtained, if we have knowledge about the entire feature-target distribution $P_{\mathbf{X},Y}$
- Assume $(\mathbf{X}, Y) \in \mathbb{R}^{d+1}$ is a jointly distributed continuous random vector, such that $P_{\mathbf{X},Y}$ is determined by a density function $p(\mathbf{x}, y)$ on $\mathbb{R}^{d+1}$.
- $\mathbf{X}$ is not random in all contexts.
- E.g. X_i could be months of a year.

Optimal Regression

- **Representation Theorem:** If $E[|Y|] < \infty$, there is a function $f: \mathbb{R}^d \rightarrow \mathbb{R}$ and a random variable ε such that:

$$Y = f(\mathbf{X}) + \varepsilon,$$

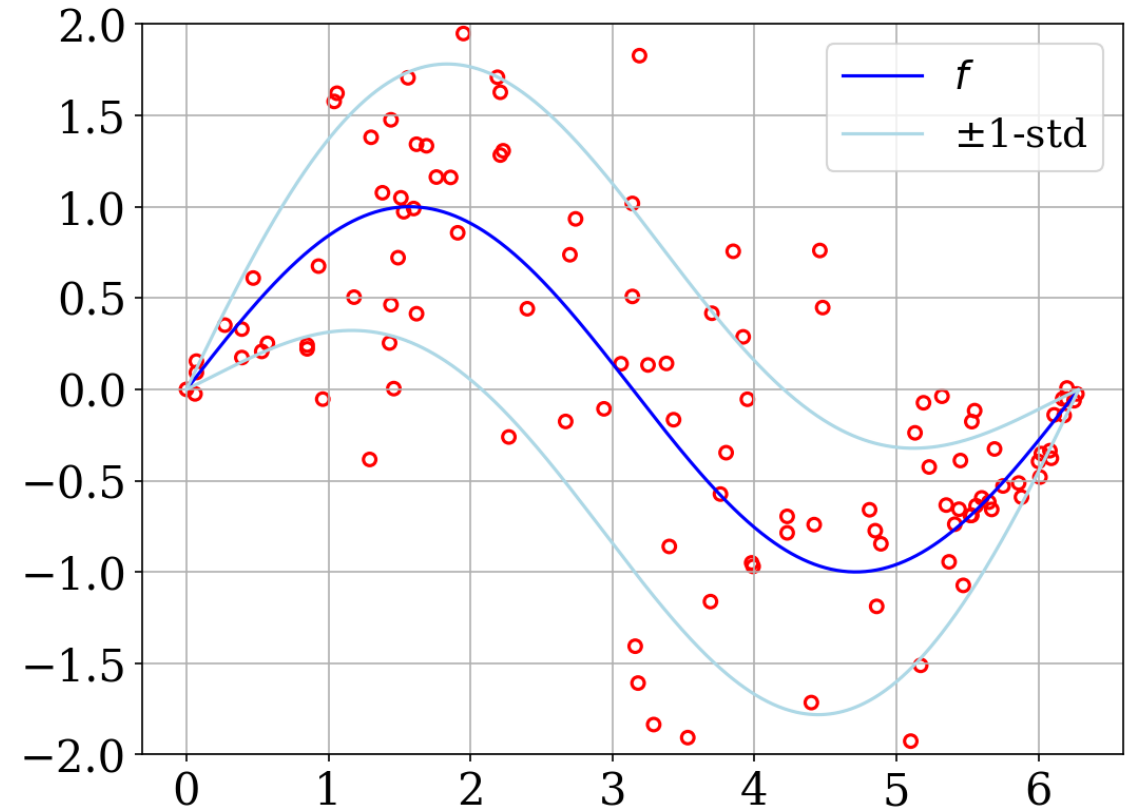
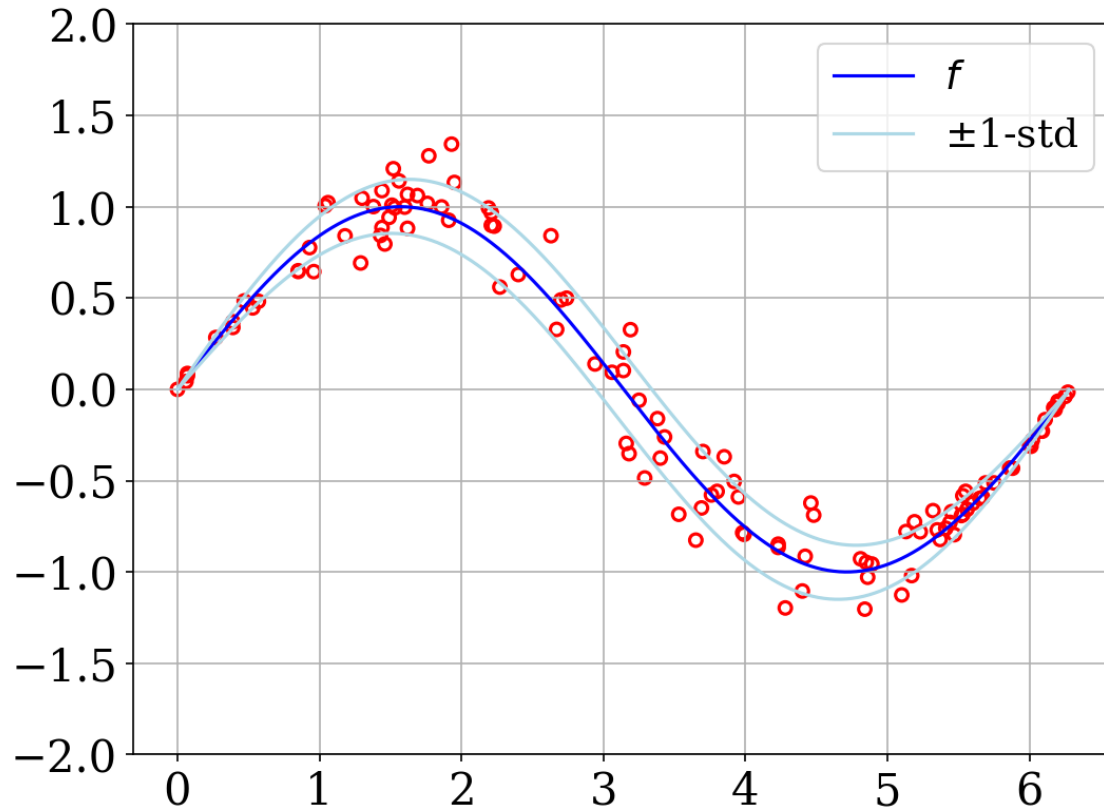
where $E[\varepsilon | \mathbf{X}] = 0$.

- The target can be decomposed as the sum of a function and a zero-mean additive noise term.
- An optimal regression function f^* satisfies

$$f^* = \arg \min_{f \in F} L[f],$$

where F is the set of all functions on $\mathbb{R}^d$ and L is the regression error specified by a loss function.

Sinusoidal Regression



- f is a good predictor of Y , when the noise is small.

Optimal Regression

- Common loss functions are:
 - Quadratic loss: $\ell(y, f(\mathbf{x})) = (y - f(\mathbf{x}))^2$,
 - Absolute loss: $\ell(y, f(\mathbf{x})) = |y - f(\mathbf{x})|$,
 - Minkowski loss: $\ell(y, f(\mathbf{x})) = |y - f(\mathbf{x})|^q, \forall q > 0$,
- The optimal regression error is $L^* = L(f^*)$.
- Most common loss function is quadratic loss or squared error loss.
- Absolute loss is more immune to outliers.
- Minkowski loss is a family of function which includes both quadratic and absolute loss as special cases.

Sample-based Regression

- In practice, the joint distribution $P_{\mathbf{X},Y}$ is not fully known, we work with an estimated f_n obtained from S_n .

- The sample-based MSE of f_n is

$$L_n = L[f_n] = E \left[(Y - f_n(\mathbf{X}))^2 \mid S_n \right]$$

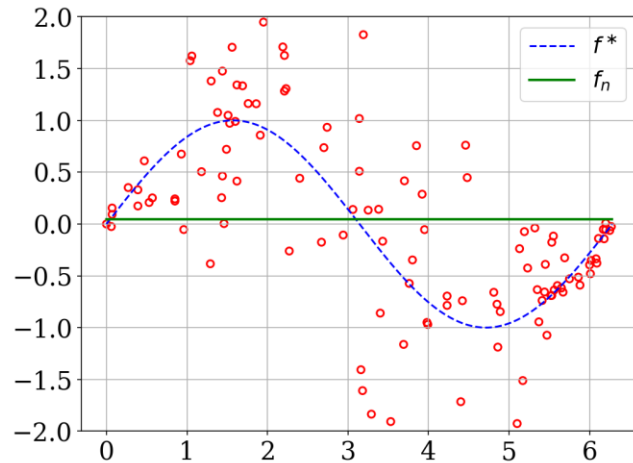
- Since the data is random, f_n and L_n are random. $E[L_n]$ gives a data-independent expected MSE.

$$\begin{aligned} E[L_n] &= L^* + E[\bar{L}[f_n]] = L^* + E \left[(f^*(\mathbf{X}) - f_n(\mathbf{X}))^2 \right] \\ &= L^* + (E[f^*(\mathbf{X}) - f_n(\mathbf{X})])^2 + \text{Var}(f^*(\mathbf{X}) - f_n(\mathbf{X})) \\ &= L^* + \text{Bias}(f_n)^2 + \text{Variance}(f_n) \end{aligned}$$

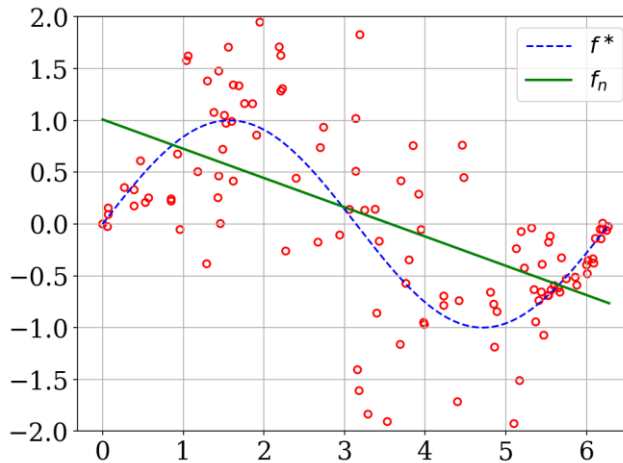
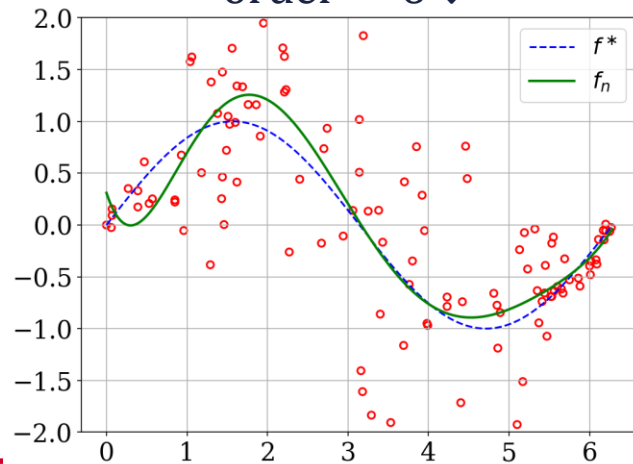
Bias-Variance Trade-Off

- The bias term explains how far off f_n is from f^* .
- The variance term measures the sensitivity of f_n to data: small if f_n changes little with different data, large if not.
- We need both bias and variance to be as small as possible.
- However small bias is associated with large variance and vice-versa.
- Hence, we have to deal with bias-variance tradeoff.
- This tradeoff can be seen in the series of regression results in the next slide.
- We use least squares polynomial regression to illustrate this:
 - At low order, there is underfitting, with large bias.
 - At very high order, there is obvious overfitting and a large variance.

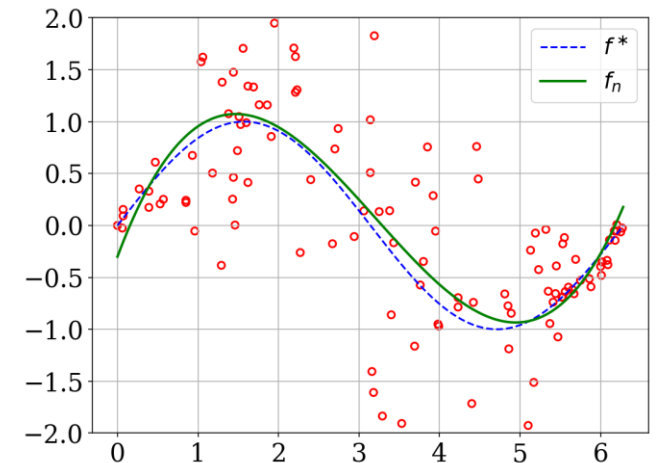
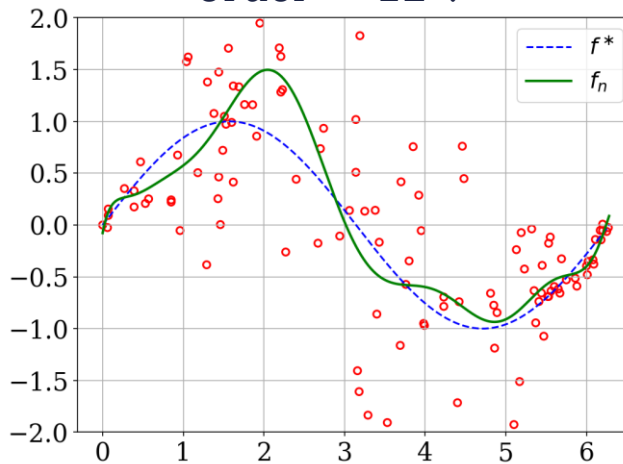
Bias-Variance Trade-Off



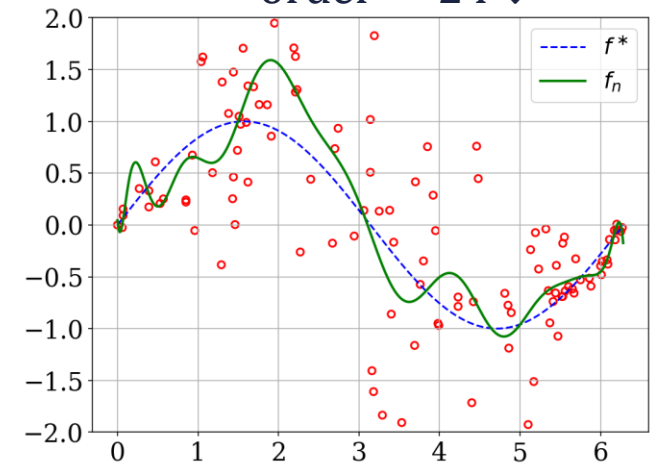
order = 0 $\uparrow$
order = 6 $\downarrow$



order = 1 $\uparrow$
order = 12 $\downarrow$



order = 3 $\uparrow$
order = 24 $\downarrow$



Parametric Regression

- In parametric regression, the representation theorem takes the form:

$$Y = f(\mathbf{X}; \boldsymbol{\theta}) + \varepsilon,$$

where $\boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^m$ is a parameter vector.

- To find a sample-based estimate $\hat{\boldsymbol{\theta}}_n$ such that the plug-in sample-based regression $f_n(x) = f(x; \boldsymbol{\theta}_n)$ has small error.
- Given $S_n = \{(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n)\}$, the least-squares estimator of parametric regression is given by:

$$\hat{\boldsymbol{\theta}}_n^{LS} = \arg \min_{\boldsymbol{\theta} \in \Theta} \sum_{i=1}^n (Y_i - f(X_i; \boldsymbol{\theta}))^2$$

- The quantity minimized is known as **residual sum of squares (RSS)**.

Linear Regression

- Simplest parametric regression model:

$$Y = a_0 + a_1X_1 + \cdots + a_dX_d + \varepsilon$$

with $X = (X_1, \dots, X_d) \in \mathbb{R}^d$ and $\boldsymbol{\theta} = (a_0, a_1, \dots, a_d) \in \mathbb{R}^{d+1}$.

- This model can be extended to basis-function linear regression model:

$$Y = \theta_0\phi_0(\mathbf{X}) + \theta_1\phi_1(\mathbf{X}) + \cdots + \theta_k\phi_k(\mathbf{X}) + \varepsilon = \boldsymbol{\Phi}(\mathbf{X})^T \boldsymbol{\theta} + \varepsilon$$

where $\phi_i: \mathbb{R}^d \rightarrow \mathbb{R}$ are basis functions, $\boldsymbol{\Phi} = (\phi_0, \phi_1, \dots, \phi_k)$, $\boldsymbol{\theta} = (\theta_0, \theta_1, \dots, \theta_k) \in \mathbb{R}^{k+1}$

- $k + 1$ is the order of the model.
- Basis functions are general and need not be linear. The model is linear in the parameters.

Linear Regression

$$Y = \theta_0\phi_0(\mathbf{X}) + \theta_1\phi_1(\mathbf{X}) + \dots + \theta_k\phi_k(\mathbf{X}) + \varepsilon = \mathbf{\Phi}(\mathbf{X})^T \boldsymbol{\theta} + \varepsilon$$

- With $\phi_0(\mathbf{X}) = 1$, $\phi_1(\mathbf{X}) = X_1, \dots, \phi_d(\mathbf{X}) = X_d$, $\theta_i = a_i, \forall i = 0, 1, \dots, d$, the above becomes the standard linear regression.
- Here $k = d$ but basis function need not be related to dimensionality of data.
- An example of basis-function linear regression is **polynomial regression**.
In the univariate case:

$$Y = a_0 + a_1X + a_2X^2 + \dots + a_kX^k + \varepsilon$$

where the basis functions are $\phi_0(X) = 1$, $\phi_1(X) = X$, $\phi_2(X) = X^2, \dots, \phi_k(X) = X^k$, $\theta_i = a_i$ for $i = 0, 1, \dots, k$.

Linear Regression

- Given the sample, $S_n = \{(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n)\}$,

$$Y_1 = \theta_0 \phi_0(\mathbf{X}_1) + \theta_1 \phi_1(\mathbf{X}_1) + \dots + \theta_k \phi_k(\mathbf{X}_1) + \varepsilon_1$$

$$Y_2 = \theta_0 \phi_0(\mathbf{X}_2) + \theta_1 \phi_1(\mathbf{X}_2) + \dots + \theta_k \phi_k(\mathbf{X}_2) + \varepsilon_2$$

$$\vdots$$

$$Y_n = \theta_0 \phi_0(\mathbf{X}_n) + \theta_1 \phi_1(\mathbf{X}_n) + \dots + \theta_k \phi_k(\mathbf{X}_n) + \varepsilon_n$$

- In matrix form:

$$\mathbf{Y}_{n \times 1} = \mathbf{H}_{n \times (k+1)}(\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n) \boldsymbol{\theta}_{(k+1) \times 1} + \boldsymbol{\varepsilon}_{n \times 1}$$

where $\mathbf{Y} = (Y_1, Y_2, \dots, Y_n)$, $\boldsymbol{\theta} = (\theta_0, \theta_1, \dots, \theta_k)$, $\boldsymbol{\varepsilon} = (\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n)$, and

$$\mathbf{H}(\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n) = \begin{bmatrix} \phi_0(\mathbf{X}_1) & \dots & \phi_k(\mathbf{X}_1) \\ \vdots & \ddots & \vdots \\ \phi_0(\mathbf{X}_n) & \dots & \phi_k(\mathbf{X}_n) \end{bmatrix}$$

- Thus, the least squares estimator can be written as:

$$\hat{\boldsymbol{\theta}}_n^{LS} = \arg \min_{\boldsymbol{\theta} \in \Theta} \|\mathbf{Y} - \mathbf{H}\boldsymbol{\theta}\|^2$$

Ridge Regression

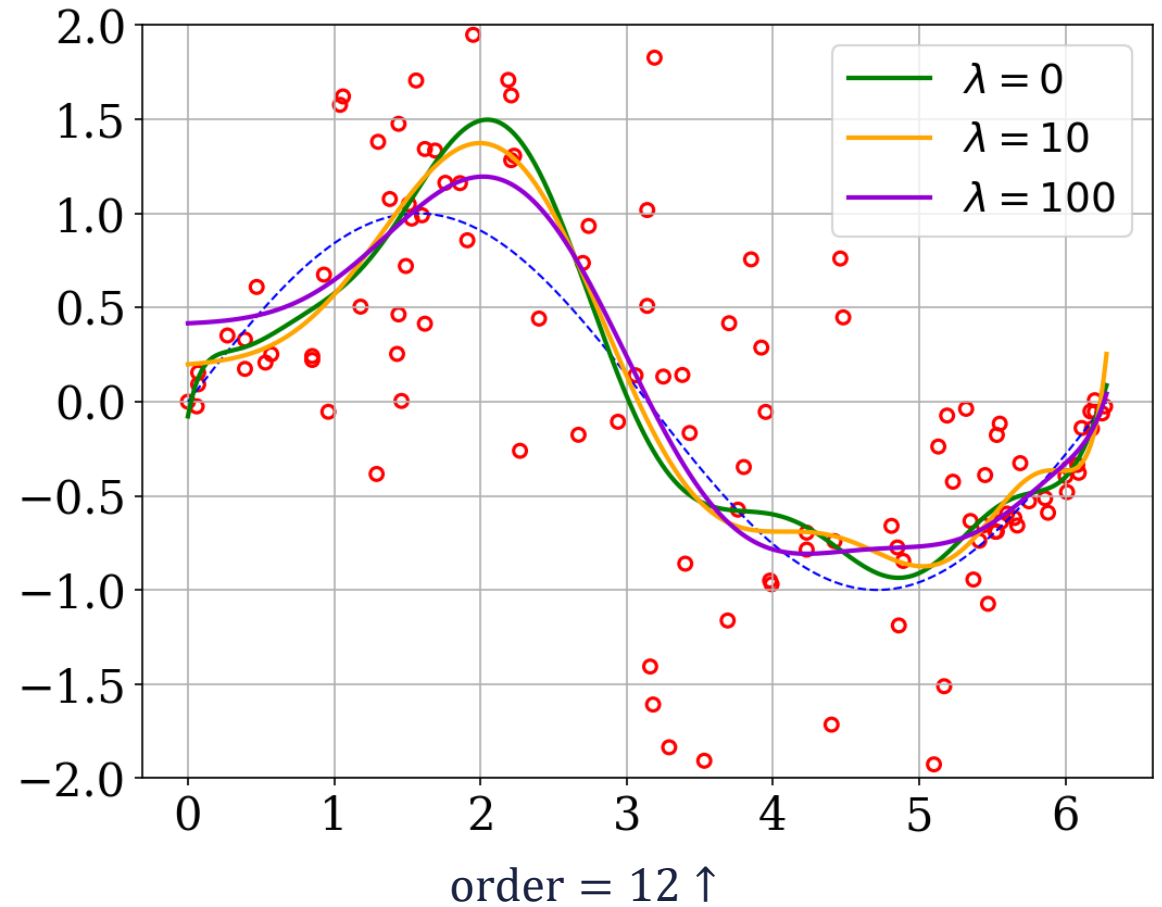
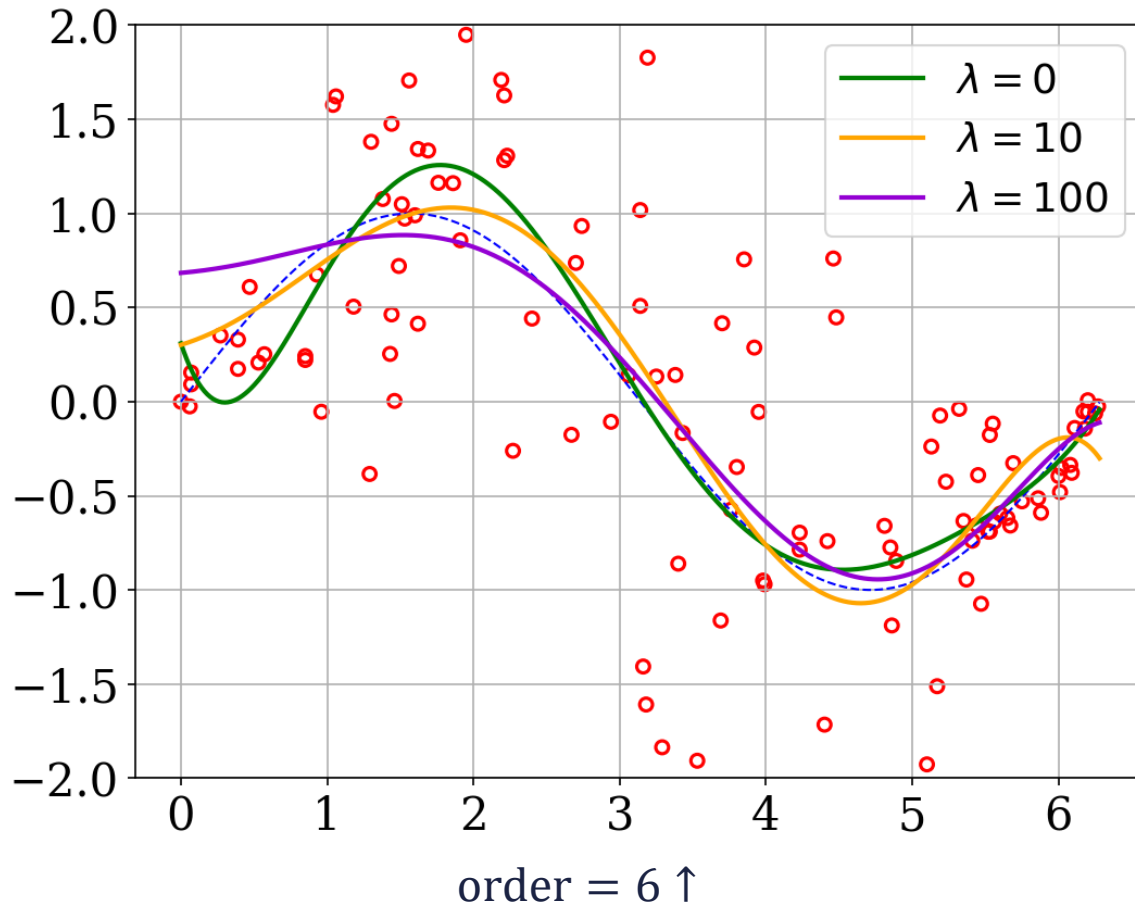
- In some cases it is desirable to constraint the parameters of the parametric regression models to avoid overfitting.
- Known as **penalized least squares** or **ridge regression**:

$$\hat{\boldsymbol{\theta}}_n^{\text{RIDGE}} = \arg \min_{\boldsymbol{\theta} \in \Theta} \|\mathbf{Y} - \mathbf{H}\boldsymbol{\theta}\|^2 + \lambda \|\boldsymbol{\theta}\|^2$$

where $\lambda > 0$ is a tunable parameter.

- $\lambda \|\boldsymbol{\theta}\|^2$ forces the parameter vector to have a small magnitude.
- In the case $\lambda = 0$, this reduces to ordinary least-squares regression.
- Larger λ forces parameters closer to zero, this process is called shrinkage.
- But the parameters never quite become zero.

Ridge Regression



Larger λ value produce flatter curves corresponding to shrinkage of polynomial coefficients to zero.

Other variants

- If the L^2 norm in ridge regression is replaced by the L^1 norm, we get the **Least Absolute Shrinkage and Selection Operator (LASSO)** estimator:

$$\hat{\boldsymbol{\theta}}_n^{\text{LASSO}} = \arg \min_{\boldsymbol{\theta} \in \Theta} \|\mathbf{Y} - \mathbf{H}\boldsymbol{\theta}\|^2 + \lambda \|\boldsymbol{\theta}\|_1$$

where $\|\boldsymbol{\theta}\|_1 = |\theta_1| + |\theta_2| + \dots + |\theta_k|$ is the L^1 norm of $\boldsymbol{\theta}$.

- Unlike ridge regression, LASSO can drive the coefficient values to zero, producing sparse solution.
- **Elastic Net** is another kind of penalized least-squares estimator combining L^1 and L^2 penalty terms

$$\hat{\boldsymbol{\theta}}_n^{\text{ENet}} = \arg \min_{\boldsymbol{\theta} \in \Theta} \|\mathbf{Y} - \mathbf{H}\boldsymbol{\theta}\|^2 + \lambda_1 \|\boldsymbol{\theta}\|_1 + \lambda_2 \|\boldsymbol{\theta}\|^2$$

where $\lambda_1, \lambda_2 > 0$.

- Produces a result intermediate between ridge and LASSO regression.

Nonparametric Regression

- Recalling Representation Theorem :

$$Y = f(\mathbf{X}) + \varepsilon$$

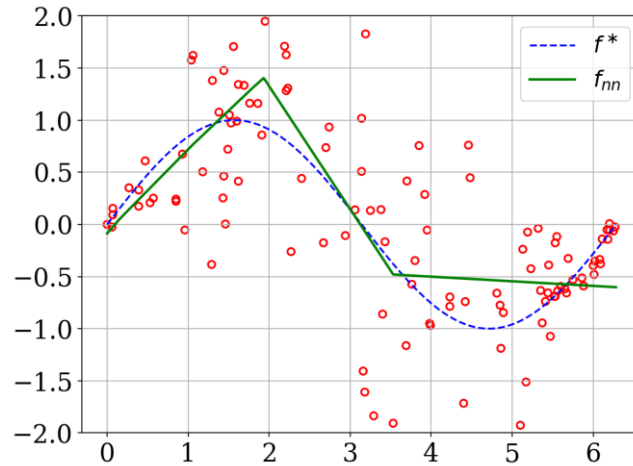
- We can directly estimate the function f as was the case with classification.
- The idea is to smooth a function to the training data.
- Two widely used methods for nonparametric regression:
 - Kernel Regression
 - Gaussian Process

Function Approximation Regression

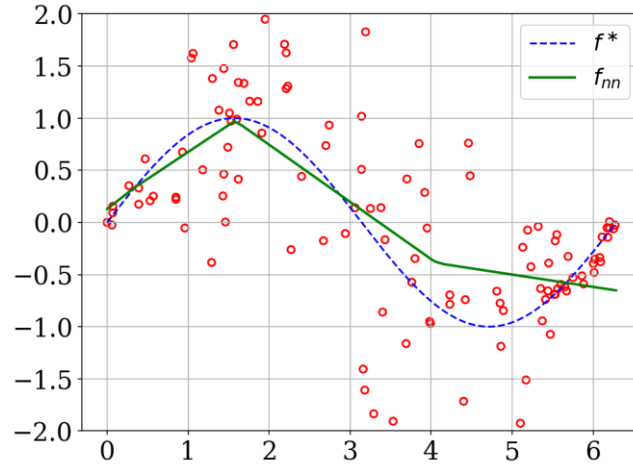
- Similarly, most of the function-approximation classification rules discussed can be modified for regression.
- **SVM Regression:**
 - The points are supposed to be within a margin around the regression line.
 - Slack variables are associated to outliers that break the margin criterion.
 - Only support vectors determine regression curve.
 - Nonlinear regression can be achieved by kernel trick as before.
- **Neural Network Regression:**
 - For example, removing the output nonlinearity, we get a neural network regressor.
 - Error criterion can be squared difference between network output and target.
 - Training works as before.

Neural Network Regression

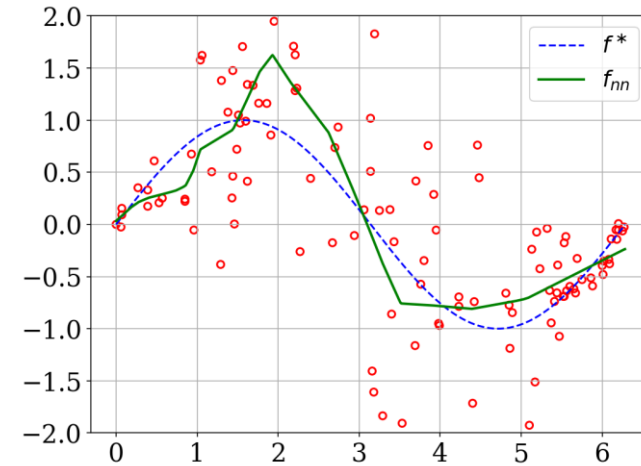
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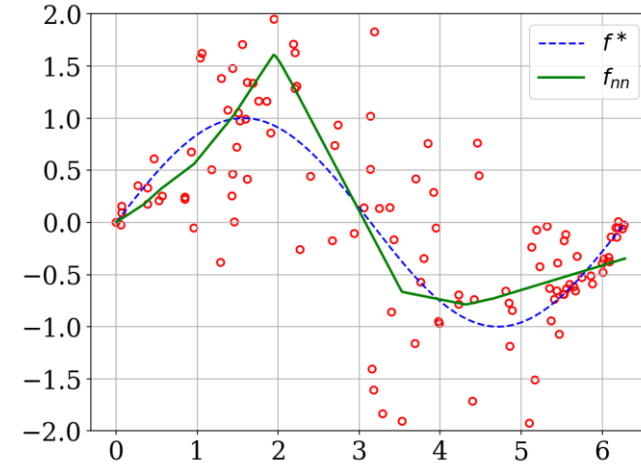
layers = 2 →



← layers = 4



← layers = 8

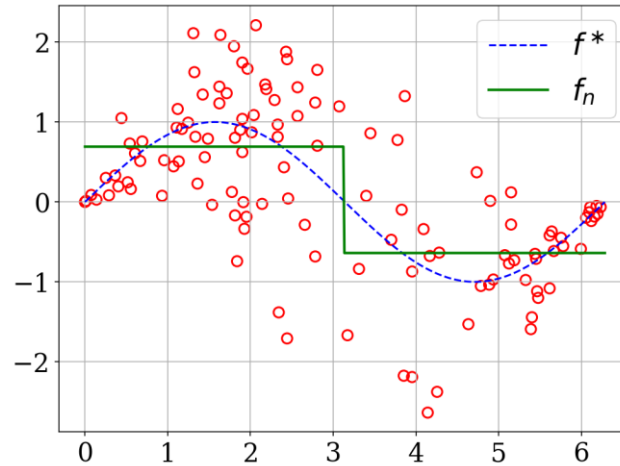


CART Regression

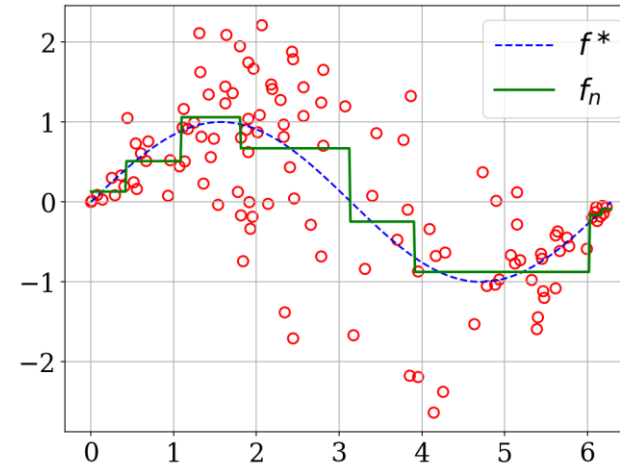
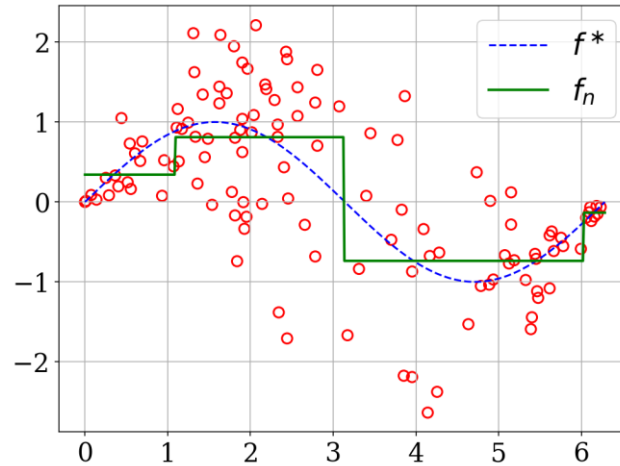
- In the case of CART, node splitting proceeds as before.
- Each node is assigned the average value of training points inside it.
- The best splits are those that minimize the residual sum of squares (RSS) between the average values and the training values in each nodes.
- Overfitting is quite common, which warrants regularization techniques like stopped splitting or pruning.

CART Regression

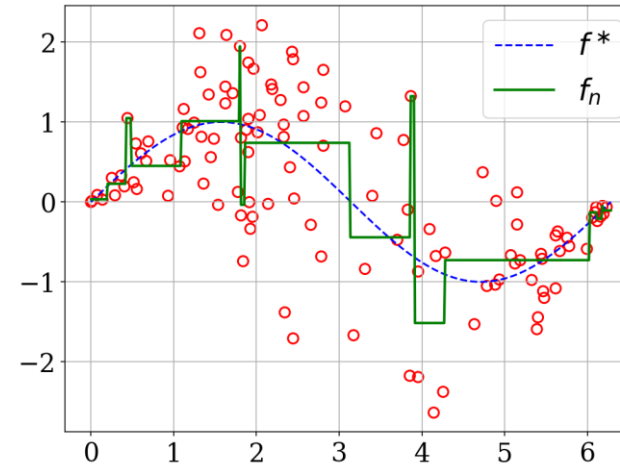
depth = 1 →



depth = 2 →



← depth = 3



← depth = 4

Error Estimation for Regression

- Given the sample, $S_m = \{(\mathbf{X}_i^t, Y_i^t); i = 1, \dots, m\}$, not used in training, regression test-set error estimator is given as:

$$\hat{L}_{n,m} = \frac{1}{m} \sum_{i=1}^m (Y_i^t - f_n(\mathbf{X}_i^t))^2$$

- Given the training data S_n , $\hat{L}_{n,m} \rightarrow L_n$ with probability 1 as $m \rightarrow \infty$, regardless of $P_{\mathbf{X},Y}$.
- Satisfactory performance of test-set error estimator depends on having large enough train, and test sets.
- Otherwise we use resubstitution error estimation:

$$\hat{L}_n = \frac{1}{n} \sum_{i=1}^n (Y_i - f_n(\mathbf{X}_i))^2$$

- This is the normalized residual sum of squares (RSS) and tends to be optimistically biased.

R^2 Statistic

- The RSS depends on the range of the targets.
- An alternative for regression is the **R^2 Statistic**.
- Let $\bar{Y}$ be an estimator of Y such that:

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

- The residual sum of squares (RSS) of the above estimator is:

$$\text{TSS} = \sum_{i=1}^n (Y_i - \bar{Y})^2$$

where TSS stands for **total sum of squares**.

R^2 Statistic

- The R^2 Statistic shows the relative improvement in RSS by using X to predict Y over not using it.

$$R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

- R^2 Statistic is also known as the **coefficient of determination**.
- A higher R^2 Statistic is more desirable and varies between 0 and 1.

Feature Selection

- Dimensionality reduction is needed for improving prediction accuracy and reducing computational costs.
- Three ways to perform feature selection for regression:
 - Wrapper Search
 - Statistical testing of coefficients
 - Sparsification methods
- **Wrapper Search:**
 - Analogous to classification.
 - An empirical criterion is defined, e.g. the resubstitution estimate, or R^2 score.
 - The best features can be searched exhaustively or greedily.
 - Greedy search can employ sequential forward/backward, or floating searches.

Wrapper Search

- R^2 score should be used carefully as generally it goes up with number of features added.
- To avoid this, we can use **adjusted R^2 statistic**:

$$\text{Adjusted } R^2 = 1 - \frac{\text{RSS} / (n - d - 1)}{\text{TSS} / (n - 1)}$$

- This will penalize models with too many variables relative to sample size.
- Another penalized criterion to use is **Mallows' C_p** . In the case of least-squares regression, this is given by:

$$C_p = \frac{\text{RSS}}{n} + \frac{2d}{n} \hat{\sigma}^2$$

Where $\hat{\sigma}$ is the MLE variance estimate.

Statistical Testing

- Statistical testing of regression coefficients can be done to reduce the number of variables.
- E.g. In linear regression,
 - test the hypothesis that each coefficient θ_i is non-zero.
 - Reject those variables for which the hypothesis cannot be rejected.
- This can be done using stepwise search algorithms, including backward elimination.
- However, when the number of parameters are high, this causes issues.

Sparsification Methods

- Seeks to shrink the model coefficients to zero, generating sparse feature vectors.
- What are some sparsification method that we saw already?

- LASSO: Adds an L1 penalty to shrink coefficients, zeroing out irrelevant features.

$$\hat{\boldsymbol{\theta}}_n^{\text{LASSO}} = \arg \min_{\boldsymbol{\theta} \in \Theta} \|\mathbf{Y} - \mathbf{H}\boldsymbol{\theta}\|^2 + \lambda \|\boldsymbol{\theta}\|_1$$

- Elastic Net: Combines LASSO's sparsity with Ridge's stability, preventing over-shrinkage of correlated features.

$$\hat{\boldsymbol{\theta}}_n^{\text{ENet}} = \arg \min_{\boldsymbol{\theta} \in \Theta} \|\mathbf{Y} - \mathbf{H}\boldsymbol{\theta}\|^2 + \lambda_1 \|\boldsymbol{\theta}\|_1 + \lambda_2 \|\boldsymbol{\theta}\|^2$$

- They inherently perform feature selection as part of their optimization process.
- Suppose you have three features: x_1 , x_2 (highly correlated), and x_3 (noise).
 - LASSO: Might discard x_1 or x_2 randomly and possibly keep x_3 .
 - Elastic Net: Likely retains both x_1 and x_2 (with similar coefficients) and discards x_3 .

Thank You